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SELECTED DATA SCIENCE PROJECTS

M.S. Capstone: Groundwater Lithium & Dementia Prevalence | *MSDS Best Capstone Presentation*

Tools: Python, Jupyter, MySQL, pandas, NumPy, GeoPandas, statsmodels, Census TIGER/Line

- Merged 460,137 groundwater records with dementia prevalence across 254 Texas counties; engineered exposure features and a homogeneity framework, using OLS with HC3 errors, permutation tests, bootstrap CIs, Cook's D, and leave-one-out analysis alongside choropleth mapping.

Applied ML for Conversion Modeling

Tools: Python, scikit-learn, gradient boosting, PySpark, Databricks, Snowflake SQL, Git

- Applied classification modeling to a business conversion problem in an insurance setting — comparing algorithms and interpreting key drivers to support prioritization and marketing decisions.

Diabetes Readmission Triage & Capacity Planning

Tools: Python, pandas, NumPy, scikit-learn, Jupyter, matplotlib, seaborn

- Built a leakage-controlled risk-ranking pipeline across 71,518 hospital encounters to prioritize patients for limited transitional-care capacity; benchmarked with recall-at-capacity and lift, then translated results into staffing scenarios and stakeholder recommendations.

LLM/RAG Document Q&A Application

Tools: Python, Databricks, PyMuPDF, sentence-transformers, Hugging Face Transformers, FAISS, PyTorch, NLTK, OpenAI API, Gradio

- Built an end-to-end application for PDF ingestion, text chunking, embeddings, vector retrieval, and source-grounded answer generation, with an interactive interface and secure API-key management through Databricks Secrets.

Texas Groundwater Quality Database & Geospatial Analysis

Tools: MySQL, SQL, Python, pandas, GeoPandas, Folium, Jupyter, VS Code

- Designed a 64-table, third-normal-form database for 20+ years of Texas groundwater data across 254 counties; standardized units, built multi-table aggregation queries and ETL workflows, and created interactive choropleth maps of mineral concentrations.

Unsupervised Learning: Clustering & Association Rule Mining

Tools: Python, scikit-learn, pandas, NumPy, matplotlib, seaborn, mlxtend, Jupyter

- Compared six clustering algorithms on 100,000 health records using internal and external validation metrics with PCA projections; mined 541,000 retail transactions with Apriori and FP-Growth, identifying 79 rules with lift up to 24.

Time Series Forecasting: Eli Lilly (LLY)

Tools: R, RStudio, tswge, ARMA, ARIMA, VAR, MLP neural network

- Compared forecasting approaches for monthly LLY log price across quarterly and annual horizons. ARIMA performed best at three months, while an MLP using a lagged competitor signal was strongest at twelve months — outperforming the sector ETF.

Heart Disease Prediction

Tools: R, RStudio, GLMNET, logistic regression, KNN, LDA, QDA, ROC/AUC

- Compared classification models on 300,000+ CDC BRFSS records. A GLMNET-selected non-additive logistic model performed best (AUROC 0.845); documented class-imbalance limitations and interpreted the strongest health predictors.

Employee Attrition Prediction: Frito-Lay

Tools: R, RStudio, caret, KNN, Naive Bayes, ggplot2, cost-based thresholding

- Tuned the classification threshold to minimize expected turnover cost rather than maximize accuracy. The selected KNN model reached 94.29% sensitivity at 72.76% accuracy and translated predictions into retention-cost scenarios.

Hospital Length of Stay Prediction

Tools: R, RStudio, multiple linear regression, LASSO, KNN, bootstrap CIs

- Compared an additive regression, a LASSO-guided model with nonlinear and interaction terms, and KNN across 113 hospitals. The complex regression performed best ($R^2 = 0.776$), and infection risk remained significant across specifications.

Sale Price Prediction (Ames Housing)

Tools: R, RStudio, multiple linear regression, LASSO, Kaggle, regression diagnostics

- Modeled Ames housing prices using 1,460 homes and 79 variables; tested neighborhood interactions and validated assumptions with Q-Q, leverage, Cook's D, and Durbin-Watson diagnostics, achieving adjusted $R^2 = 0.867$ and Kaggle RMSLE = 0.150.

Crab Age Prediction & Interactive App

Tools: R, RStudio, R Shiny, linear regression, feature engineering, model validation

- Engineered polynomial, shell-density, and size-by-sex interaction features to predict crab age. The final model achieved $R^2 = 0.702$ and MAE = 1.14 months, with an R Shiny app for interactive exploration.